Project Report

Healthcare: Be Healthy

AI/ML-Based Wellness Dashboard

This report presents the design, implementation, features, and working of an interactive health prediction and wellness recommendation system.

1. Introduction

Healthcare: Be Healthy is an AI/ML-based wellness dashboard developed to help users assess their health condition through simple lifestyle and biometric inputs. The tool is designed to provide meaningful insights such as health score, organ risk estimation, life expectancy prediction, and personalized health suggestions.

The main purpose of this project is to spread health awareness in an easy, interactive, and engaging way. Instead of only displaying numbers, the dashboard helps users understand how daily habits like sleep, smoking, alcohol consumption, calorie intake, and exercise can influence their overall well-being.

2. Problem Statement

Many people do not regularly track their health until visible problems appear. Traditional health monitoring tools are often either too clinical or not interactive enough for everyday users. There is a need for a simple and accessible system that can estimate wellness, highlight risk factors, and motivate healthier lifestyle choices.

3. Objectives

- To develop an interactive health assessment dashboard.
- To predict an overall health score using machine learning.
- To estimate lungs and liver risk based on lifestyle habits.
- To project life expectancy using user-provided health indicators.
- To analyze the impact of habits using frequency-based comparisons.
- To generate personalized recommendations for improving health.

4. Scope of the Project

This project is designed as a wellness awareness and educational support tool. It can be useful for students, fitness beginners, and general users who want to understand the effect of their lifestyle choices. The project does not replace a doctor or medical diagnosis, but it acts as an informative decision-support and awareness system.

5. Features of the System

- Interactive sliders for user input
- Overall Health Score prediction (0–100)
- Lungs and Liver Risk Assessment
- Life Expectancy Prediction
- Frequency-based habit impact analysis
- Personalized health recommendations
- Graphs, charts, and motivational insights

6. Technologies Used

Component	Technology / Library
Programming Language	Python
Machine Learning Model	Random Forest Regressor
Feature Scaling	MinMaxScaler
Data Handling	NumPy, Pandas
Visualization	Matplotlib
Interactive UI	ipywidgets
Notebook Environment	Jupyter Notebook / Google Colab

7. Methodology

The project follows a data-driven machine learning workflow consisting of data simulation, preprocessing, model training, prediction, and dashboard visualization.

7.1 Data Simulation

A synthetic dataset of 500 users is created to simulate realistic health and lifestyle patterns. The dataset includes variables such as age, weight, height, smoking, alcohol use, exercise duration, sleep hours, calorie intake, and caffeine consumption.

A custom health score is generated using weighted factors such as BMI, age influence, physical activity, and harmful lifestyle habits. This synthetic score is then used as the target variable for machine learning prediction.

7.2 Data Preprocessing

The collected input features are normalized using MinMaxScaler. Scaling ensures that all features contribute proportionally during model training and prediction.

7.3 Machine Learning Model

A Random Forest Regressor is used to predict the overall health score. This model was chosen because it is robust, handles non-linear relationships effectively, and performs well on structured tabular data.

7.4 Prediction Logic

Based on user input, the system predicts:

- Overall Health Score
- Lungs Risk
- Liver Risk
- Life Expectancy

7.5 Frequency-Based Analysis

One of the unique features of the project is its ability to compare the impact of user habits depending on their frequency. For example, smoking or alcohol consumption can be analyzed as daily, weekly, or occasional behavior to show how frequency affects health outcomes.

8. Working of the System

- The user enters biometric and lifestyle details through interactive sliders.
- The system processes the data and applies scaling.
- The trained machine learning model predicts the health score.
- Rule-based logic estimates lungs and liver risk.
- A life expectancy approximation is calculated.
- The dashboard displays the results with charts, summary cards, and recommendations.

9. Input Parameters

- Age
- Weight
- Height
- Smoking habit
- Alcohol consumption
- Exercise duration/frequency
- Sleep hours
- Daily calorie intake
- Caffeine intake

10. Output Parameters

- Overall Health Score (0–100)
- Lungs Risk Level
- Liver Risk Level
- Estimated Life Expectancy
- Habit Frequency Impact
- Personalized Recommendations

11. Recommendation System

The recommendation system provides personalized suggestions according to the user's lifestyle profile. For example, if the user reports poor sleep, the dashboard may suggest improving sleep duration and consistency. If the BMI is high or exercise is low, the system may recommend physical activity and dietary balance.

- Smoking reduction or avoidance
- Alcohol moderation
- Improved sleep schedule
- Regular exercise
- Healthy calorie intake
- Weight and BMI management

12. Advantages of the Project

- Easy to use and beginner-friendly
- Interactive and visually engaging
- Provides instant health-related insights
- Combines machine learning with practical recommendations
- Useful for awareness, education, and self-monitoring

13. Limitations

- Uses synthetic data instead of real clinical records
- Predictions are approximate and not medically certified
- Life expectancy and organ risk are estimations, not diagnoses

- Performance depends on the assumptions used while generating health scores

14. Future Enhancements

- Use real-world healthcare datasets for better accuracy
- Add more organs and disease risk modules
- Deploy the project as a web application using Streamlit or Flask
- Include user login and progress tracking
- Integrate wearable or fitness band data
- Add report export and personalized trend monitoring

15. Conclusion

Healthcare: Be Healthy is a meaningful AI/ML-based project that combines health awareness with interactive analytics. It demonstrates how machine learning can be used to estimate wellness, evaluate risks, and encourage healthier habits in a simple and user-friendly way.

The project is especially valuable as an educational and portfolio-ready application because it blends machine learning, data visualization, and user interaction into a practical real-world solution.

16. References

- Python Documentation
- Scikit-learn Documentation
- Matplotlib Documentation
- Pandas Documentation
- NumPy Documentation
- ipywidgets Documentation